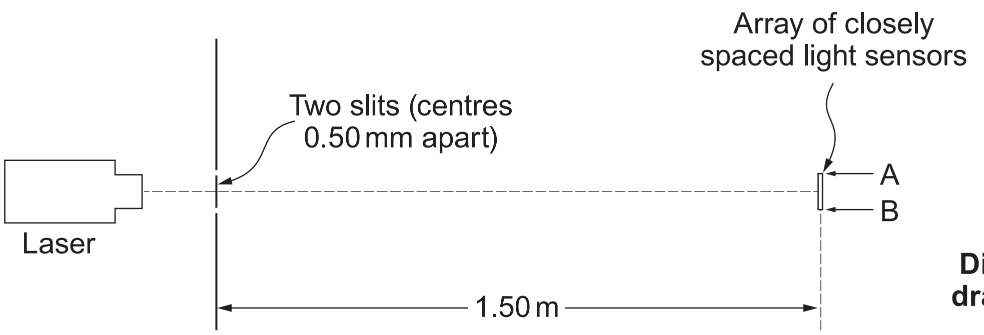
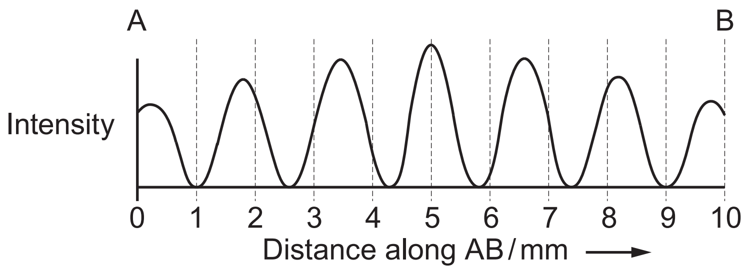
Examiner only

4. (a) A modern version of Young's double slit experiment is set up as shown.



Diagrams not drawn to scale

The array of light sensors is connected to circuitry that displays a graph of light intensity along the line AB.



(i) Calculate a value for the wavelength of light from the laser. [3]

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(ii) Each slit is several wavelengths wide. Suggest why the intensity of the bright fringes decreases the further the fringes are from the centre of the pattern. [2]

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Examiner only

(iii) A student suggests that replacing the laser with one that emits near infra-red (that is infra-red just beyond the end of the visible spectrum) will increase the fringe separation.

I. Explain whether or not the student is right. [2]

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II. State **one** way in which the fringe separation could be increased without changing the laser. [1]

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(b) More than 200 years ago Thomas Young drew a conclusion from his ‘fringes’ experiment. It was not generally accepted for several years. State what conclusion Young drew, **and** suggest what needed to be done by the scientific community for the conclusion to be accepted. [3]

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